



# Proposed CEA Link — December 2026

A 24-line link built from the December 2026 staffing rules with no duty over 8h40, judged on the ORR fatigue factors — nothing carried over from today's roster except the rules

Prepared 12 September 2026 · re-rendered 24 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylebone Roster v24.22) · figures on this page are computed, not typed

## PROPOSAL

# Eight Forty

By the Book, with no duty over 8h40

CODE **EF-24-E21**

FINGERPRINT **Ocf19f56**

BUILT FROM duty table E · seed 21 · 24 lines · 4 cover weeks

LINEAGE Family EF — "By the Book" (BB-24-D7 · 0f14abce) under one more rule: no duty runs over 8h40. Five of that table's eleven Mon–Sat turns did (its earlies ran to 9h30), so the whole December 2026 duty table was searched again under the same rules with the ceiling at 8h40, and the rotation then searched for the ORR factors exactly as "By the Book" was. Nothing is carried over from today's roster. Compare with "By the Book", "Same Turns" (ST-24-B7 · d15e1b74) and "Quarter To" (QT-24-Q34 · 70cf9874).

✓ **24** lines designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

✓ **0** fatigue factors present (today: 5)

## 26 turns

in the December 2026 duty table, capped

searched against the timetable under the 8h40 rule; lates shorter than earlies; the longest duty is 8h 40m

## 6 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

## 0 of 25

fatigue factors present

today's link has 5; every other factor is clear or does not apply

## 6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20)

## 4 cover weeks

lines 1, 7, 13, 19

evenly spaced six lines apart — today's sit at 1, 7, 12, 17

## 1h 31m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

## What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 — *By the Book* under one more rule: **no duty runs over 8h40**. That rule is not a trim. Five of *By the Book*'s eleven Monday-to-Saturday turns ran past it (its earlies reached 9h30), and a day's minutes are fixed by the contract — 14 duties paying 7,000 minutes is a mean of 8h20 — so with the ceiling twenty minutes above the mean, every duty has to sit between 7h and 8h40 and most of them near the top. The December 2026 duty table was therefore searched again from nothing under the same rules: four to open every day, three through to the close and four on a Saturday, five still on at 22:00, fourteen on a Saturday and ten on a Sunday, late turns shorter than early ones, no :05 or :10 times, at most two people on a turn, and the times set against the measured December 2026 service. It pays the 35-hour week to the minute across 20 working lines and four cover weeks.

The rotation was then searched for one thing: the rules it will be assessed against. Chiltern's 13-day limit, twelve hours between duties and the exact contract were treated as walls; the 25 ORR good-practice fatigue factors were the objective, each one scored by the workspace's own assessFatigue on every one of about a million candidate arrangements, together with the size of the week-to-week step, weekends off and an even spread of cover weeks. Only after every factor was settled was a week's coherence allowed to count — one turn a week where it costs nothing against the factors — so that a rotation is not chopped up for no fatigue gain. The rules-only version, with no such term, is on page 8 for comparison. Page 8 says how, and how to reproduce it.

**What it is not.** Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 6 and 7 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

BEFORE THE FIGURES

# The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

## The question

Should the link for the December 2026 timetable be built from the December 2026 timetable rules and the measured timetable, accepting shift times nobody works today, or keep today's times?

**Eight Forty** is a rules-first table under the 8h40 cap. It is read against **Quarter To (QT-24-Q34)**, the strongest case for keeping today's times — not against every design in the folder.

## The criteria, in three tiers

**HARD — A DESIGN MEETS THESE OR CANNOT BE RUN**

- ✓ Twelve hours between duties — met, tightest 14h 20m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

**SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER**

- December 2026 headcounts met: **4 of 4** (today 0 of 4)
- Fatigue factors present: **0** (today 5)
- Demand fit weekday · Sat · Sun: **70.4** · 16.2 · 20.9 (today 44.7 · 45.7 · 65.9; lower is closer)
- Times worked today: **2 of 26** · full weekends off: **6** in 24 (today 4 in 20)

**PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S**

- Early against late: earlies 7h 15m–8h 40m, lates 7h 25m–8h 40m
- 24 of 26 clock times are new — the duty table on page 4 lists them
- The longest duty, 8h 40m, and who holds it
- Cover weeks at lines 1, 7, 13, 19

## How much each figure can bear

| FIGURE ON THESE PAGES                                       | WHERE IT COMES FROM                                                                          | WEIGHT                                      |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------|
| 0 rests under 12h · 35h 00m contract · 14/10 at the weekend | Computed from the cells; re-computed by the workspace on import                              | <b>Firm</b>                                 |
| Longest run 6 days, inside 13                               | Computed; the 13-day limit's policy citation is still outstanding                            | <b>Firm figure</b> , unconfirmed threshold  |
| Four at the open, three to close, 14 and 10 at the weekend  | Owner-relayed December 2026 headcounts with no document behind them (class C)                | <b>Provisional</b>                          |
| 0 fatigue factors present                                   | The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm" | <b>Advisory only</b> — never pass or fail   |
| Demand fit 70.4 · 16.2 · 20.9                               | Measured timetable, one share-based measure on one December 2026 timetable curve             | <b>Indicative</b>                           |
| 2 of 26 times are today's                                   | Computed against the live 20-line link                                                       | <b>Firm</b> , but a proxy for acceptability |
| Search score 10710                                          | The annealer's own objective; page 8 explains it                                             | <b>Do not weigh</b>                         |

## What no page here can tell you

- Whether the people who would work them accept a table in which 24 of 26 times are new (page 4 lists them) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (9 as a block, 11 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 13, 19); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.

## Before it is frozen

Lines 2, 4, 5, 6, 8 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 7 mean anything, and only then is this sheet more than an aid to a conversation.

## THE ROTATION

## The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

| LINE    | SUN            | MON            | TUE            | WED            | THU            | FRI            | SAT            | MON-SAT<br>hours | ALL WEEK<br>hours | DAYS WORKED<br>Mon-Sat | TURNS<br>Mon-Fri |
|---------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|-------------------|------------------------|------------------|
| 1       | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      | —                |
| 2       | 07:15<br>15:30 | RD             | RD             | 06:20<br>14:50 | 06:20<br>14:50 | 06:20<br>14:50 | 06:20<br>15:00 | 34h 10m          | 42h 25m           | 4                      | <b>1</b>         |
| 3       | RD             | RD             | 06:20<br>14:00 | 06:20<br>14:00 | 06:20<br>14:00 | 06:20<br>14:00 | RD             | 30h 40m          | 30h 40m           | 4                      | <b>1</b>         |
| 4       | RD             | 07:20<br>16:00 | 06:20<br>15:00 | 06:20<br>15:00 | 06:20<br>15:00 | 06:20<br>15:00 | RD             | 43h 20m          | 43h 20m           | 5                      | 2                |
| 5       | RD             | 09:20<br>18:00 | 09:20<br>18:00 | 09:20<br>18:00 | 09:20<br>18:00 | RD             | RD             | 34h 40m          | 34h 40m           | 4                      | <b>1</b>         |
| 6       | RD             | 09:20<br>17:55 | RD             | 14:55<br>22:40 | RD             | 09:20<br>17:55 | 09:20<br>18:00 | 33h 35m          | 33h 35m           | 4                      | 2                |
| 7       | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      | —                |
| 8       | 16:00<br>23:25 | RD             | RD             | 14:55<br>22:40 | 14:55<br>22:40 | 15:30<br>23:55 | 15:35<br>23:55 | 32h 15m          | 39h 40m           | 4                      | 2                |
| 9       | RD             | RD             | 09:20<br>17:55 | 09:20<br>17:55 | 09:20<br>17:55 | 09:20<br>18:00 | RD             | 34h 25m          | 34h 25m           | 4                      | 2                |
| 10      | RD             | 07:20<br>16:00 | 07:20<br>16:00 | 07:20<br>16:00 | 07:20<br>16:00 | RD             | 06:20<br>14:55 | 43h 15m          | 43h 15m           | 5                      | <b>1</b>         |
| 11      | 07:15<br>15:50 | 06:20<br>14:55 | 06:20<br>14:55 | 06:20<br>14:55 | RD             | RD             | 06:20<br>14:50 | 34h 15m          | 42h 50m           | 4                      | <b>1</b>         |
| 12      | 07:15<br>14:30 | 06:20<br>14:50 | 06:20<br>14:50 | RD             | RD             | 07:20<br>16:00 | 07:00<br>15:30 | 34h 10m          | 41h 25m           | 4                      | 2                |
| 13      | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      | —                |
| 14      | 09:15<br>17:55 | 09:20<br>18:00 | RD             | RD             | 15:45<br>23:55 | 15:45<br>23:55 | 15:30<br>23:55 | 33h 25m          | 42h 05m           | 4                      | 2                |
| 15      | 15:50<br>23:25 | 14:55<br>22:40 | 14:55<br>22:40 | RD             | RD             | 14:55<br>22:40 | 14:00<br>22:00 | 31h 15m          | 38h 50m           | 4                      | <b>1</b>         |
| 16      | 14:25<br>22:25 | 16:00<br>23:55 | 16:00<br>23:55 | 16:00<br>23:55 | RD             | RD             | 14:45<br>22:45 | 31h 45m          | 39h 45m           | 4                      | <b>1</b>         |
| 17      | 15:30<br>23:25 | 15:45<br>23:55 | 15:45<br>23:55 | 15:45<br>23:55 | 14:55<br>22:40 | RD             | RD             | 32h 15m          | 40h 10m           | 4                      | 2                |
| 18      | RD             | 15:30<br>23:55 | 15:30<br>23:55 | 15:30<br>23:55 | 15:30<br>23:55 | RD             | 15:40<br>23:55 | 41h 55m          | 41h 55m           | 5                      | <b>1</b>         |
| 19      | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      | —                |
| 20      | 07:15<br>15:55 | 06:20<br>15:00 | RD             | RD             | 06:20<br>14:55 | 06:20<br>14:55 | 06:20<br>14:15 | 33h 45m          | 42h 25m           | 4                      | 2                |
| 21      | RD             | RD             | 07:20<br>16:00 | 07:20<br>16:00 | 07:20<br>16:00 | 07:20<br>16:00 | 09:20<br>18:00 | 43h 20m          | 43h 20m           | 5                      | 1                |
| 22      | RD             | RD             | 09:20<br>18:00 | 09:20<br>18:00 | 09:20<br>18:00 | 09:20<br>18:00 | RD             | 34h 40m          | 34h 40m           | 4                      | <b>1</b>         |
| 23      | RD             | 06:20<br>14:00 | RD             | RD             | 16:00<br>23:55 | 16:00<br>23:55 | 15:45<br>23:55 | 31h 40m          | 31h 40m           | 4                      | 2                |
| 24      | 14:25<br>22:25 | 14:55<br>22:40 | 14:55<br>22:40 | RD             | RD             | 14:55<br>22:40 | 14:00<br>22:00 | 31h 15m          | 39h 15m           | 4                      | <b>1</b>         |
| Cover   | 10<br>SP 4     | 14<br>SP 4     | 14<br>SP 4     | 14<br>SP 4     | 14<br>SP 4     | 14<br>SP 4     | 14<br>SP 4     | 35h 00m          | 38h 21m           | 4.20                   | 10<br>one turn   |
| Minutes | 4,820          | 7,000          | 7,000          | 7,000          | 7,000          | 7,000          | 7,000          | 42,000           | 46,820            |                        |                  |

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

# How a week compares with today

Nothing here was asked to resemble today's roster; this is how close the fatigue factors took it on their own, and where the December 2026 duty table differs

## The shape of a week, against today

| MEASURE                                                  | TODAY (20)          | PROPOSED (24)      |
|----------------------------------------------------------|---------------------|--------------------|
| Working weeks that are one turn (one clock time Mon–Fri) | 7 of 16             | 10 of 20           |
| Weeks mixing early and late turns                        | 7                   | 2                  |
| Rest-day breaks that are two days or more                | 13 of 17            | 20 of 24           |
| Single rest days between duties (not a 48h break)        | 4                   | 4                  |
| Days worked in a week (lines × days)                     | 1×3, 3×4, 11×5, 1×6 | 6×4, 14×5          |
| Cover (spare) weeks (whole weeks)                        | lines 1, 7, 12, 17  | lines 1, 7, 13, 19 |
| Distinct shift times (2 on today's roster)               | 18                  | 26                 |
| Mon–Sat hours, average week (the contract)               | 35h 00m             | 35h 00m            |
| Sunday duties (not contracted; half the working lines)   | 8 of 16 lines       | 10 of 20 lines     |

## Changed — the December 2026 headcount

| PEOPLE ON DUTY                                            | TODAY     | PROPOSED  | WHY                                                             |
|-----------------------------------------------------------|-----------|-----------|-----------------------------------------------------------------|
| Monday to Friday                                          | 11–12     | 14        | the contract: 20 working lines × 35h has to be worked somewhere |
| Saturday                                                  | 10        | 14        | owner's figure for December 2026, leaning late for events       |
| Sunday                                                    | 8         | 10        | owner's figure for December 2026                                |
| Opening at 06:20 (07:15 Sunday) wk · Sat · Sun            | 4 · 4 · 3 | 4 · 4 · 4 | four to open, every day — met                                   |
| Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun | 2 · 2 · 3 | 3 · 4 · 3 | three to close, four on a Saturday — met                        |
| Still on duty at 22:00 wk · Sat · Sun                     | 4 · 2 · 3 | 5 · 5 · 5 | five from 22:00, Monday to Saturday — met                       |

## The duty table — the December 2026 timetable rules under 8h40, beside today's

| DECEMBER 2026 TURN         | WK | SAT | SUN |
|----------------------------|----|-----|-----|
| 06:20–14:00 7h 40m · Early | 1  |     |     |
| 06:20–14:15 7h 55m · Early |    | 1   |     |
| 06:20–14:50 8h 30m · Early | 1  | 1   |     |
| 06:20–14:55 8h 35m · Early | 1  | 1   |     |
| 06:20–15:00 8h 40m · Early | 1  | 1   |     |
| 07:00–15:30 8h 30m · Early |    | 1   |     |
| 07:15–14:30 7h 15m · Early |    |     | 1   |
| 07:15–15:30 8h 15m · Early |    |     | 1   |
| 07:15–15:50 8h 35m · Early |    |     | 1   |
| 07:15–15:55 8h 40m · Early |    |     | 1   |
| 07:20–16:00 8h 40m · Early | 2  |     |     |
| 09:15–17:55 8h 40m · Early |    |     | 1   |
| 09:20–17:55 8h 35m · Early | 1  |     |     |
| 09:20–18:00 8h 40m · Early | 2  | 2   |     |
| 14:00–22:00 8h 00m · Late  |    | 2   |     |
| 14:25–22:25 8h 00m · Late  |    |     | 2   |
| 14:45–22:45 8h 00m · Late  |    | 1   |     |
| 14:55–22:40 7h 45m · Late  | 2  |     |     |
| 15:30–23:25 7h 55m · Late  |    |     | 1   |
| 15:30–23:55 8h 25m · Late  | 1  | 1   |     |
| 15:35–23:55 8h 20m · Late  |    | 1   |     |
| 15:40–23:55 8h 15m · Late  |    | 1   |     |
| 15:45–23:55 8h 10m · Late  | 1  | 1   |     |
| 15:50–23:25 7h 35m · Late  |    |     | 1   |
| 16:00–23:25 7h 25m · Late  |    |     | 1   |
| 16:00–23:55 7h 55m · Late  | 1  |     |     |

Read the table in three blocks: 6 turns Monday to Saturday alike (and 5 the weekday keeps to itself), Saturday's own 6, and Sunday's 9 for its 07:15–23:25 window. Openers stagger by their finish and closers by their start so nobody hands over a cliff. 19 of the 35 start and finish instances sit on :00, :15, :30 or :45; the 12 that do not (14:50, 14:55, 07:20, 09:20, 17:55, 22:40, 15:35, 15:40, 15:50, 15:55, 14:25, 22:25) are what four distinct opener finishes cost under a cap that leaves the long earlies about twenty minutes of room — *By the Book* has three such times.

| TODAY'S TURN (NONE IS REUSED) | WK | SAT | SUN |
|-------------------------------|----|-----|-----|
| 06:20–13:35 7h 15m            | 1  |     |     |
| 06:20–13:45 7h 25m            | 1  |     |     |
| 06:20–14:00 7h 40m            |    | 1   |     |
| 06:20–14:20 8h 00m            | 2  |     |     |
| 06:20–14:50 8h 30m            |    | 3   |     |
| 07:15–15:45 8h 30m            |    |     | 3   |
| 08:00–16:30 8h 30m            | 2  |     |     |
| 08:30–16:30 8h 00m            |    |     | 1   |
| 11:00–19:30 8h 30m            | 1  |     |     |
| 12:00–20:00 8h 00m            |    | 1   |     |
| 13:00–21:00 8h 00m            |    |     | 1   |
| 13:30–21:00 7h 30m            |    | 1   |     |
| 13:30–22:00 8h 30m            | 1  |     |     |
| 14:00–22:30 8h 30m            | 2  |     |     |
| 14:30–22:00 7h 30m            |    | 2   |     |
| 14:30–23:25 8h 55m            |    |     | 3   |
| 14:45–23:55 9h 10m            |    | 2   |     |
| 15:15–23:55 8h 40m            | 2  |     |     |

06:20–14:00 is the one short early; every other early (8h 30m–8h 40m) is longer than every late (7h 45m–8h 25m) — the lever against unpopular lates, at the only size the cap allows: the shortest long early is 5 minutes longer than the longest late, where *By the Book* had fifteen and an hour on average. 2 of the 26 turns are times people work today (06:20–14:00, 06:20–14:50) — the same trade as *By the Book*, and the question to put to the room rather than to the tool.

# People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

## Monday to Friday window 06:20–23:55 · one row per distinct weekday

|                       | 05 | 06 | 07  | 08  | 09  | 10 | 11 | 12 | 13 | 14 | 15 | 16  | 17  | 18  | 19  | 20 | 21 | 22 | 23 | FIT  |
|-----------------------|----|----|-----|-----|-----|----|----|----|----|----|----|-----|-----|-----|-----|----|----|----|----|------|
| TODAY MON · WED       |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 7  | 7  | 7  | 7   | 5   | 5   | 5   | 4  | 4  | 4  | 2  | 55.8 |
| TODAY TUE · THU · FRI |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 8  | 8  | 8  | 8   | 6   | 6   | 6   | 5  | 5  | 4  | 2  | 50   |
| PROPOSED              |    | 4  | 6   | 6   | 9   | 9  | 9  | 9  | 9  | 10 | 9  | 8   | 8   | 5   | 5   | 5  | 5  | 5  | 3  | 70.4 |
| DEC 2026 TRAFFIC      | 5  | 69 | 107 | 127 | 122 | 95 | 85 | 75 | 84 | 79 | 87 | 105 | 140 | 138 | 107 | 90 | 73 | 75 | 66 |      |

## Saturday window 06:20–23:55

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY            |    | 4  | 4  | 4  | 4  | 4  | 4  | 5  | 6  | 9  | 6  | 6  | 6  | 6  | 6  | 5  | 4  | 2  | 2  | 45.7 |
| PROPOSED         |    | 4  | 5  | 5  | 7  | 7  | 7  | 7  | 7  | 10 | 10 | 9  | 9  | 7  | 7  | 7  | 7  | 5  | 4  | 16.2 |
| DEC 2026 TRAFFIC | 6  | 24 | 58 | 75 | 81 | 92 | 88 | 77 | 64 | 68 | 72 | 84 | 82 | 67 | 64 | 79 | 78 | 62 | 41 |      |

## Sunday window 07:15–23:25

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY            |    |    | 3  | 4  | 4  | 4  | 4  | 4  | 5  | 8  | 8  | 5  | 4  | 4  | 4  | 4  | 3  | 3  | 3  | 65.9 |
| PROPOSED         |    |    | 4  | 4  | 5  | 5  | 5  | 5  | 5  | 7  | 8  | 6  | 6  | 5  | 5  | 5  | 5  | 5  | 3  | 20.9 |
| DEC 2026 TRAFFIC |    |    | 15 | 45 | 65 | 71 | 74 | 71 | 61 | 55 | 64 | 69 | 78 | 73 | 75 | 52 | 71 | 65 | 50 |      |

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window — lower is better, and today's link scores what it scores. It is measured on duty minutes, so an hour a duty only touches counts in proportion; the cells above count heads. It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 8. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish (the last at 23:54, three of them arrivals) — the standing question on whether Sunday's window moves; the window is stored per design, so the proposal can be rebuilt to either answer.

**Reading it.** The table's times were searched against this curve on a five-minute grid, so the cover follows the traffic as closely as the day's fixed minutes allow. The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

# The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

**Hard limits** — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)  
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.  
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)  
Tightest anywhere: **14h 20m — line 4 Mon 07:20-16:00 into Tue 06:20-15:00** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks  
700h of duty a week across 20 working lines. Sundays (80.33h) sit on top as RDW, as they do today. Individual weeks range 30h 40m–43h 20m; only the average is the contract.

**December 2026 timetable design figures** — the staffing shape agreed for the new timetable

| RULE                                           | PROPOSAL                                                                                             |                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Four on at the open, every day                 | ✓ 4 weekday · 4 Saturday · 4 Sunday                                                                  |                                                                    |
| Three through to the close; four on a Saturday | ✓ 3 weekday · 4 Saturday · 3 Sunday                                                                  |                                                                    |
| Five still on duty at 22:00                    | ✓ 5 weekday · 5 Saturday                                                                             | a 22:00 finish is not "on at 22:00"                                |
| Fourteen on a Saturday, ten on a Sunday        | ✓ 14 · 10                                                                                            |                                                                    |
| Four cover weeks, evenly spread                | ✓ lines 1, 7, 13, 19 — gaps 6, 6, 6, 6                                                               | whole weeks, never scattered days                                  |
| About 4.2 days a week worked, Mon-Sat          | ✓ 4.20 over the 20 working lines                                                                     |                                                                    |
| No :05 or :10 times except the open and close  | ✓ none                                                                                               | today's times already obey this                                    |
| Late turns slightly shorter than most earlies  | ✓ met on every day — lates 7h25–8h25; one short early a day (7h15–7h55), every other early 8h15–8h40 | one short open turn, then every other early longer than every late |
| No duty over 8h40                              | ✓ longest 8h40                                                                                       | every day class — By the Book's longest is 9h30, today's 9h10      |

## To settle before it is frozen

**The size of the lever.** Met, and the cap has squeezed it: 14 duties paying 7,000 minutes average 8h20, so with nothing over 8h40 the long earlies sit at 8h30–8h40 and the lates at 7h45–8h25 — the shortest long early is 5 minutes longer than the longest late, against *By the Book's* 15. Whether that still buys anything against unpopular lates is the room's question. And on a Saturday the earlies can average at most 24 minutes longer than the lates (*By the Book* asks 30; four closers with distinct starts leave one split that pays 7,000): this table has 20, the weekday 31.1, Sunday 30. **The clock.** 12 times sit off the quarter hour against *By the Book's* three — the price of four distinct opener finishes when the cap leaves the long earlies twenty minutes of room. **Sunday's minutes** are not fixed by the contract; the search took the largest Sunday total at which every pin holds — 4,820, a mean of 8h02 against *By the Book's* 4,965. **Familiarity.** 2 of the 26 turns are times people work today (06:20-14:00, 06:20-14:50) — the same trade as *By the Book*. **Sunday's finish** — five December 2026 timetable movements fall after 23:25; the proposal inherits today's window deliberately rather than deciding it.

# ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021).  $\triangle$  present ·  $\checkmark$  clear ·  $\bullet$  standing, a property of the operation · — not applicable. Nathan assesses against this list.

| CODE | FACTOR                                                                                | TODAY'S LINK                                        | PROPOSED                                             |
|------|---------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| FF1  | Night shift covering 00:00–05:00 Time of day                                          | — —                                                 | — —                                                  |
| FF2  | Early shift starting 05:00–07:00 Time of day                                          | $\bullet$ 24                                        | $\bullet$ 24                                         |
| FF3  | Very early shift starting before 05:00 Time of day                                    | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF4  | Very early shift (before 05:00) over 8h Duty length                                   | — —                                                 | — —                                                  |
| FF5  | Day shift over 12h Duty length                                                        | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF7  | Early shift over 10h Duty length                                                      | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF8b | Fewer than 2 days rest after a block of early starts Recovery time                    | $\triangle$ 3                                       | $\checkmark$ 0                                       |
| FF11 | More than 13 consecutive shifts without a 48h break Recovery time                     | $\checkmark$ 12<br>$\checkmark$ 9 as rostered       | $\checkmark$ 11<br>$\checkmark$ 9 as rostered        |
| FF10 | More than 4 consecutive 12h day shifts Cumulative                                     | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF15 | More than 4 consecutive early shifts in a rotating pattern Cumulative                 | $\triangle$ 6                                       | $\checkmark$ 4                                       |
| MRSF | More than 12 consecutive day shifts Cumulative                                        | $\checkmark$ 9                                      | $\checkmark$ 6                                       |
| MRSF | More than 55 hours worked in any 7-day period Cumulative                              | $\triangle$ 60.8                                    | $\checkmark$ 51.8                                    |
| MRSF | More than 7 consecutive 8h shifts (definition to confirm) Cumulative                  | $\triangle$ 9                                       | $\checkmark$ 6                                       |
| MRSF | More than 6 consecutive night or early shifts in a permanent pattern Cumulative       | — —                                                 | — —                                                  |
| FF17 | Backward rotating pattern (definition to confirm) Circadian                           | $\checkmark$ 10 backward / 12 forward               | $\checkmark$ 12 backward / 12 forward                |
| FF18 | Rotating pattern of about a week (definition to confirm) Circadian                    | $\bullet$ 20-line rotation · typically 4h 0m a week | $\bullet$ 24-line rotation · typically 1h 31m a week |
| FF19 | Successive start times varying by more than 2 hours (definition to confirm) Circadian | $\triangle$ 6                                       | $\checkmark$ 0                                       |
| FF6  | Night shift over 10h Night shifts                                                     | — —                                                 | — —                                                  |
| FF8  | Less than 2 days rest after consecutive nights Night shifts                           | — —                                                 | — —                                                  |
| FF9  | Less than 14h rest in any 24h for night shifts Night shifts                           | — —                                                 | — —                                                  |
| FF12 | Only one day rest after night shifts Night shifts                                     | — —                                                 | — —                                                  |
| FF14 | More than 4 consecutive nights in a rotating pattern Night shifts                     | — —                                                 | — —                                                  |
| FF16 | More than 3 consecutive night shifts over 8h Night shifts                             | — —                                                 | — —                                                  |
| FF20 | First night shift Night shifts                                                        | — —                                                 | — —                                                  |
| FF13 | Less than 12h rest in any 24h Intervals between duties                                | $\checkmark$ 0                                      | $\checkmark$ 0                                       |

Present: 0 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

# How it was chosen

Reproducible: the same table, seed and objective give the same grid on any machine

## How it was chosen

**1 • The table.** The December 2026 timetable rules re-solved under the cap (table-book.mjs). Every duty LENGTH structure that pays a day exactly — four openers with distinct finishes, three closers with distinct starts (four on a Saturday), the middles, one short early below every late and every other early above it, nothing over 8h40 — was enumerated on the five-minute grid: 540,737 for a weekday, 84,116 for a Saturday, 1,479,340 for a Sunday (whose minutes the contract does not fix: the search walked down from the week's 8h20 mean to the largest total at which every pin holds, 4,820). For each, the free start times were searched on the quarter hour for the closest fit to the December 2026 timetable demand curve under every other pin of the default table's test — no :05 or :10, at most two on a turn, starts and finishes a quarter apart, five on at 22:00, the busiest hour never on a trough, Saturday's evening lean, and no more clock times than *By the Book* asks a reader to hold. The placement is a role-preserving anneal over the whole day — every move keeps the day's minutes — and the winner's lengths are checked against the enumeration; the best fit won, with the quarter hour preferred and Saturday asked to reuse the weekday's turns where that cost nothing. Two pins gave way to arithmetic and are on page 6: Saturday's half-hour mean gap, and the quarter-hour spacing of opener finishes.

**2 • The grid.** Simulated annealing over the 24-line grid: 100,000 candidate arrangements per restart, five restarts from different random starting points per run, four seeded runs — over a million candidates in all — then a steepest-descent polish of every same-day swap and every pair of lines. Every move keeps each day's duties exactly as the table says — a swap between two lines on one day, or two whole lines changing places — so the coverage, the contract and the headcounts are true by construction and only the SHAPE is searched. Cover weeks stay at lines 1, 7, 13 and 19 throughout.

**3 • The judge.** Every candidate was scored by the workspace's own modules — runDesignChecks, assessFatigue, assessHardLimits, scoreOrder — with a rest under 12h or a run past six days costing more than everything else combined, then every fatigue factor present (each one, and its size: the length of an early block, the hours in seven days, each start-time jump, each backward step), then the week-to-week step and the length of a block on the same sort of week. Then, at a weight no factor can be traded for, the coherence of a week (one turn, its weekend variant) and fairness: four to five days a week, no single rest day between duties. Weekends off and long weekends were rewarded. Last, the app's line-order optimiser was offered the result and kept only where it improved that score.

## Alternatives measured

| DESIGN                                                        | RUN | FF PRESENT | WKENDS | ONE-TURN | STEP   | WK FIT | SCORE   |
|---------------------------------------------------------------|-----|------------|--------|----------|--------|--------|---------|
| EF-24-E21 · 0cf19f56 — <b>Eight Forty</b> (this proposal)     | 6   | 0          | 6      | 10/20    | 1h 31m | 70.4   | 10710   |
| EF-24-E13 · 2d7a596b — runner-up                              | 6   | 0          | 6      | 12/20    | 3h 40m | 70.4   | 18600   |
| Workspace default · 29ec9810 (Dec 2026 table, generated)      | 7   | 5          | 5      | 0/20     | 4h 16m | 30.8   | 2192436 |
| BB-24-D7 · 0f14abce — <b>By the Book</b> (a sibling proposal) | 6   | 0          | 6      | 12/20    | 2h 52m | 30.8   | —       |
| ST-24-B7 · d15e1b74 — <b>Same Turns</b> (a sibling proposal)  | 6   | 1          | 6      | 18/20    | 2h 33m | 46.3   | —       |
| QT-24-Q34 · 70cf9874 — <b>Quarter To</b> (a sibling proposal) | 6   | 0          | 6      | 19/20    | 2h 53m | 46.2   | —       |
| Today's 20-line link (for scale)                              | 9   | 5          | 4      | 7/16     | 4h 0m  | 44.7   | —       |

**How the winner was picked:** rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores). The top two tied on every rule and on fit; the search's own score decided it (10710 against 18600). The other 2 seeded results are in results/ (E34, E7); none stands higher on the pick than the rows shown. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

## LOAD IT

# The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Eight-Forty-EF-24-E21-import.txt (tab-separated, pastes directly) and Eight-Forty-EF-24-E21.json (the app's own format).

|    |             |             |             |             |             |             |             |
|----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 1  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 2  | 07:15-15:30 | RD          | RD          | 06:20-14:50 | 06:20-14:50 | 06:20-14:50 | 06:20-15:00 |
| 3  | RD          | RD          | 06:20-14:00 | 06:20-14:00 | 06:20-14:00 | 06:20-14:00 | RD          |
| 4  | RD          | 07:20-16:00 | 06:20-15:00 | 06:20-15:00 | 06:20-15:00 | 06:20-15:00 | RD          |
| 5  | RD          | 09:20-18:00 | 09:20-18:00 | 09:20-18:00 | 09:20-18:00 | RD          | RD          |
| 6  | RD          | 09:20-17:55 | RD          | 14:55-22:40 | RD          | 09:20-17:55 | 09:20-18:00 |
| 7  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 8  | 16:00-23:25 | RD          | RD          | 14:55-22:40 | 14:55-22:40 | 15:30-23:55 | 15:35-23:55 |
| 9  | RD          | RD          | 09:20-17:55 | 09:20-17:55 | 09:20-17:55 | 09:20-18:00 | RD          |
| 10 | RD          | 07:20-16:00 | 07:20-16:00 | 07:20-16:00 | 07:20-16:00 | RD          | 06:20-14:55 |
| 11 | 07:15-15:50 | 06:20-14:55 | 06:20-14:55 | 06:20-14:55 | RD          | RD          | 06:20-14:50 |
| 12 | 07:15-14:30 | 06:20-14:50 | 06:20-14:50 | RD          | RD          | 07:20-16:00 | 07:00-15:30 |
| 13 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 14 | 09:15-17:55 | 09:20-18:00 | RD          | RD          | 15:45-23:55 | 15:45-23:55 | 15:30-23:55 |
| 15 | 15:50-23:25 | 14:55-22:40 | 14:55-22:40 | RD          | RD          | 14:55-22:40 | 14:00-22:00 |
| 16 | 14:25-22:25 | 16:00-23:55 | 16:00-23:55 | 16:00-23:55 | RD          | RD          | 14:45-22:45 |
| 17 | 15:30-23:25 | 15:45-23:55 | 15:45-23:55 | 15:45-23:55 | 14:55-22:40 | RD          | RD          |
| 18 | RD          | 15:30-23:55 | 15:30-23:55 | 15:30-23:55 | 15:30-23:55 | RD          | 15:40-23:55 |
| 19 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 20 | 07:15-15:55 | 06:20-15:00 | RD          | RD          | 06:20-14:55 | 06:20-14:55 | 06:20-14:15 |
| 21 | RD          | RD          | 07:20-16:00 | 07:20-16:00 | 07:20-16:00 | 07:20-16:00 | 09:20-18:00 |
| 22 | RD          | RD          | 09:20-18:00 | 09:20-18:00 | 09:20-18:00 | 09:20-18:00 | RD          |
| 23 | RD          | 06:20-14:00 | RD          | RD          | 16:00-23:55 | 16:00-23:55 | 15:45-23:55 |
| 24 | 14:25-22:25 | 14:55-22:40 | 14:55-22:40 | RD          | RD          | 14:55-22:40 | 14:00-22:00 |

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